

Episode 01 — Answer Key (Teacher Copy)

Grade 3 · Do not distribute

The Mystery of the Magnetic Cat · FOR TEACHER USE ONLY

Section 1 — Vocabulary Matching

Magnetic Field → The invisible pushing-and-pulling zone around every magnet

Ferromagnetic → A metal naturally attracted to magnets (iron, nickel, or cobalt)

Conductor → A material that allows electricity or magnetic force to pass through it

Poles → The two ends of a magnet — North (N) and South (S)

Repel → To push away — happens when same magnetic poles face each other

Attract → To pull toward — happens when opposite magnetic poles face each other

Section 2 — Fill in the Blanks

1. Iron, Nickel, Cobalt

2. I Need Chocolate — I = Iron, N = Nickel, C = Cobalt

3. skin (also accept: hand)

4. field

5. attract

6. repel

7. iron

Section 3 — True or False

All metals are magnetic. → **FALSE**

A magnetic field can work through the air without touching an object. → **TRUE**

A copper penny will stick to a horseshoe magnet. → **FALSE**

A steel paper clip will stick to a magnet. → **TRUE**

North and South magnetic poles attract each other. → **TRUE**

North and North magnetic poles attract each other. → **FALSE**

Magnetic fields can pass through your hand to reach iron objects. → **TRUE**

Aluminum is one of the three ferromagnetic metals. → **FALSE**

Section 4 — Sort It

MAGNETIC: Steel nail, Steel paper clip, Iron fridge door

NOT MAGNETIC: Copper penny, Plastic dinosaur, Aluminum soda can, Wooden block, Rubber band

Section 5 — Short Answer (Sample Answers)

1. A magnetic field is an invisible pushing-and-pulling zone around every magnet that can attract iron objects without touching them.
2. The soda can is made of aluminum. Aluminum is NOT ferromagnetic — it does not contain iron, nickel, or cobalt — so it is not attracted to magnets.
3. I = Iron, N = Nickel, C = Cobalt. 'I Need Chocolate' helps remember the three naturally magnetic metals.
4. When two same poles are put together, they REPEL — they push away from each other. You cannot force them to touch.
5. Pip discovered that Whiskers's collar tag was made of stainless steel, which contains iron — a ferromagnetic metal. That's why the magnet pulled the collar tag.

NGSS Alignment

Standard	Description	Covered In
3-PS2-3	Ask questions to determine cause and effect relationships of electric or magnetic interactions between two objects not in contact with each other.	Sections 1, 2, 3, 5
3-PS2-4	Define a simple design problem that can be solved by applying scientific ideas about magnets.	Section 6 (Experiment)

Teacher Tips

Before watching: Ask students: 'What do you think makes something magnetic?' Record predictions. Return to them after watching.

Pause at 2:00: After nail/penny tests — ask 'What pattern do you notice about which things stick?'

Pause at 6:00: After field demo — ask 'What surprised you about the paper clip moving before being touched?'

After watching: Use Section 6 (Scavenger Hunt) as a take-home activity. Review results as a class next session.

Discussion: Ask: 'Why do you think only iron, nickel, and cobalt are magnetic?' No wrong answers — encourage curiosity.

Extension: Challenge early finishers to design a 'magnetic sorter' machine. What would it separate? How would it work?

